Enumerative Combinatorics — problem 5.26

A. Engel — Problem-Solving Strategies · 1998

Let $F(n,r)$ = number of n-permutations with exactly r cycles = Stirling number of the first kind. Prove the recurrences

$$F(n+1,r) = F(n,r-1) + nF(n,r), \qquad F(n,1) = (n-1)!, \qquad F(n,n) = 1.$$